BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

BIOLOGY

Past Paper 2023 • Paper Code: MRD-9th 160523 (A)

Retyped faithfully from the original board paper — same questions, no changes.

Section – A

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** MRD-9th 160523 (A), VERSION: C

Note: There are three sections in this paper i.e A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e A, B, C or D.

Q1. Enzyme amylase speed up the:

(A) Digestion of protein only
(B) Digestion of starch only
(C) Digestion of lipids only
(D) Digestion of cellulose only

Q2. ATP and NADPH are products of reaction in Photosynthesis:

(A) Light reaction
(B) Dark reaction
(C) Glycolysis
(D) Krebs Cycle

Q3. The rate of photosynthesis at the time of dawn and sunset is:

(A) Low
(B) High
(C) Not affected
(D) Stopped

Q4. Frequent watery, loose bowel movements and vomiting are common symptoms of:

(A) Diarrhoea
(B) Ulcers
(C) Constipation
(D) Influenza

Q5. The tough membrane that surrounds heart and protects it is:

(A) Peritoneum
(B) Pericardium
(C) Sternum
(D) Sacrum

Q6. Stem: the study of the "Dead remains of plants and animals" is:

(A) Botany
(B) Zoology
(C) Parasitology
(D) Paleontology

Q7. At various stages in stages in scientific method, biologists collect data to make hypothesis from:

(A) Observations
(B) Conclusions
(C) Experiments
(D) Deductions

Q8. Chitin in cell wall and food absorption are the characteristics refer to kingdom:

(A) Prokaryotic
(B) Protista
(C) Fungi
(D) Animalia

Q9. In animals, nerve cells contribute:

(A) Water to various parts of the body
(B) Fats to various parts of the body
(C) Vitamins to various parts of the body
(D) Coordination in the body

Q10. Ribosomes are made of almost equal amounts of proteins and ribosomal RNA. But a unique feature which allow them in prokaryotes is:

(A) Ribosomes are not bound by membrane
(B) Ribosomes are the power house of the cell
(C) Ribosomes are involved in transport materials
(D) Ribosomes provide support to the inner parts of cell

Q11. During Telophase of cell division:

(A) Spindle fibers are formed
(B) Equatorial plates are formed
(C) Nucleolus and nuclear membrane are disappeared
(D) Nucleolus and nuclear membrane are reappeared

Q12. Which one of the following statement is correct regarding prophase-II?

(A) Chromosomes arrange themselves on the equatorial plate
(B) Sister chromatids of each chromosome move to the opposite pole
(C) Sister chromatids of each chromosome reach to the opposite poles
(D) Centrioles move to opposite poles and make spindle fibers

Section – B

Time Allowed: [illegible in source scan] | **Maximum Marks:** 32

Q2. Write short answers to any eight (8) of the following parts. All parts carry equal marks:-

(i) Why biologist career as agriculturalist is important? Give FOUR reasons

(ii) With the help of diagram shows structure of virus.

(iii) Write contribution of Jean Baptist de-Lamark in cell biology.

(iv) Describe the structure of cells of epidermal tissues in plants.

(v) Define cell cycle and write names of its TWO main stages.

(vi) What are TWO differences and TWO similarities between mitosis and meiosis?

(vii) Briefly explain the mechanism of enzyme action. Also write names of two models of enzyme action.

(viii) Differentiate between TWO steps of alcoholic fermentation.

(ix) Column A contains some properties of fats. Write their answer in column B. [Table - Column A | Column B: Elements of fats | ___ ; Main sources | ___ ; Energy value | ___ ; Metabolic value | ___]

(x) Define stomata. How transpiration take place in stomata?

(xi) How oxygenated blood is transported from lung to heart?

Section – C

Maximum Marks: 21

> Note: Answer any three (3) questions. All questions carry equal marks:-

Q3.

(a) How does do scientists collect and organize their data?

(b) Enlist some serious threat faced by Indus river dolphin.

Q4.

(a) Write down any FOUR characteristics features of prokaryotic cell.

(b) With the help of late prophase diagram indicate position of spindle fiber.

Q5.

(a) How some enzymes work in complex metabolic reactions?

(b) Explain Krebs cycle with reference to its discovery, occurrence and series of reactions.

Q6.

(a) Write down any THREE advantages of inorganic fertilizers.

(b) Write any FOUR differences between vein and capillary with reference to their structure.